Manual

Statistical Process Analysis
PDV-SPA 8.2

Version 1.0.23049

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Copyright

Contents

1 Overview: Statistical Process Analysis

Overview

Purpose

This function package includes statistical reports/evaluations. Samples are generated from the raw data based on the sample parameters defined within the collection rules or their characteristics. Presenting the mean values of samples as a graphic curve over time complements the measured value curves in the graphic process analysis to show statistically smoothed data. Quality control charts enable statistical process monitoring and analyses based on process values.

Integration

This function package requires a machine interface for data collection in the HYDRA Process Communication Controller (PCC) and the licenses for process data collection and processing (PDV-PDM and PDV-VRP).

Features

Analysis/evaluation options of the statistical process analysis:

- Advanced graphic process evaluation: evaluation of samples in a correlative process analysis. Mean values of samples and process limits for samples are displayed in graphic form.
- Advanced graphic process evaluation: Samples are evaluated based on control charts using extensive filter criteria. Different control charts are supported (e.g. xq, s and R).

2 Graphic Process Analysis with Samples (based on machines)

Summary

Menu	Quality management → Process analysis → Graphic process analysis with samples (based on machines)
Transaction code	gpdasm
Function authorization	gpdasm

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measuring values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

Workplace

> The required workplace can be selected specifically in the "general" index tab using the pool of workplaces application.

Time domain from – until

> Specifies a specific period of time for the evaluation.

Consider long-term data

> Enabling this checkbox allows for long-term data to be considered.

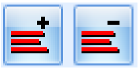
Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons  can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale“

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

3 Graphic Process Analysis with Samples (based on orders)

Summary

Menu	Quality Management → Process Analysis → Graphic Process Analysis with Samples (based on orders)
Transaction code	gpdaso
Function authorization	gpdaso

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measured values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

MES order number

Selects a specific operation (MES order number)

Batch

Selects a batch number

Time domain from – until

Selects a specific period of time

Consider long-term data

Enabling this check box allows for long-term data to be considered.

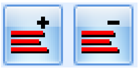
Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons  can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale“

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

4 Graphic Process Analysis with Samples (based on batches)

Summary

Menu	Quality Management → Process analysis → Graphic process analysis with samples (based on batches)
Transaction code	gpdasc
Function authorization	gpdasc

The graphic process analysis is used for the process analysis in quality management. The collection of samples enables the values to be compressed and allows for statistical preparation. The analysis function based on samples represents average values and their progression; it is an Xq chart.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. The selected process parameters can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily. In this evaluation, which refers to samples, only the averages from the samples are displayed. The collection of samples is defined in the collection request (inspection planning).

Integration

The function displays sample values from recorded measured values of process characteristics. These values must have been previously collected in the system.

Requirement

The data compression (sampling) is active as a separate process.

Selection parameters

The following selection criteria are available in the application:

Batch number

Specifically selects a produced batch.

Alternative batch number 1

Selects the alternative batch number 1 at the produced batch

Time domain from – until

Selects a period of time

Consider long-term data

Enabling this check box allows long-term data to be considered.

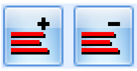
Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons  can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

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The following configuration options can be set for each chart:

Checkbox “Legend”

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Checkbox “Show scale“

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

5 Process Data Protocol

Overview

Menu	Quality management → Process analysis → Process data protocol
Transaction code	pdpu
Function authorization	pdpu

The *Process data protocol* is used for the process analysis in the quality management and in the process data processing.

Purpose

The process data protocol is a summary of recorded values to get a quick overview during the process analysis.

The table view of the detail application offers an overview of existing entries. The system sorts the displayed information using table functions and complying with specified selection parameters. Each user can change the display of fields with the context menu.

Integration

The function displays recorded measured values of process characteristics in aggregated form based on selection criteria. These values must have been previously recorded in the system.

Selection parameters

The application provides the following selection criteria:

Type of evaluation

The evaluation type defines the relating object for the summary or if the user requires a calculation for single values for all process parameters. The following relating objects can be selected:

- Order
- OP (operation)
- Machine + shift date
- Machine + shift date + shift number

Machine

The system offers a detailed search for the machine using the pool application.

Evaluation period from - to

Specifies the time interval to be selected.

Order

The system offers a detailed search for the order using the pool application. This field is mandatory if the user selects the evaluation type "Order" or "Operation".

Operation

The system offers a detailed search for the operation using the pool application. This field is mandatory if the user selects the evaluation type "OP".

Include long-term data

Long-term data is included.

Please note that the system keeps all data used for these evaluations/reports in the server memory. Consequently, we do not recommend to select long periods of time and large data sets.

Technical background info:

Every time the system requests data in the client applications, it stores this data in the server memory and only then transfers the data to the client. If you require more memory space, you need to increase the server memory reserved for the Java application accordingly. Please refer to the MPDV support.

Detail application "Process data protocol"

The tabular report *process data protocol* displays the process data recorded and saved in the database including the following information: Please note that not all columns are filled with information depending on the selected evaluation type.

Category *Statistic parameter*

Process parameter

Technical name of the recorded process parameter

Mean value (AVG - order sequencing list)

Mean value of the recorded process parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Maximum value

Maximum value of the recorded process parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Minimum value

Minimum value of the recorded parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Number of measured values

Number of recorded measured values meeting the selection criteria for the process parameter.

Category *Order*

Order

Order number for which the process parameter was entered.

Article number of order

Stored article number of the order.

Article name of order

Stored article name of the order.

Category "Operation"

Operation

MES order number for which the process parameter was recorded.

OP

OP number for which the process parameter was recorded.

OP name/designation

Stored OP designation of the operation

Article number of OP

Stored article name of the operation.

Article name of OP

Stored article name of the operation.

Category "Production parameter"

Machine

Workplace/machine number indicating where the process parameter values were recorded.

Shift date

Shift date indicating when the process parameter values were recorded.

Shift number

Shift number of the recorded process parameter values.

Category "Primary quantities"

Yield (P)

Recorded yield (primary) of the OP/machine

Scrap (P)

Recorded scrap quantity (primary) of the OP/machine

Quantity unit (P)

Quantity unit of the recorded quantity (primary)

Actual times category

Runtime

Runtime of the OP/machine

Time of production

Production time of the OP/machine

Downtime

Downtimes of the OP/machine

Category "Specification"

Target value (TV)

The target value for the collected process parameter.

Upper TL

The upper tolerance limit defined for this recorded process parameter.

UPAL (upper process action limit)

The upper process action limit defined for this recorded process parameter.

Lower TL

The lower tolerance limit defined for this recorded process parameter.

LPAL (lower process action limit)

The lower process action limit defined for this recorded process parameter.

Distribution

Graphical display of the distribution of measured values

Cp

The process capability index Cp is shown in the column Cp. The calculation of the process capability index Cp is based on the upper and lower specification limit and the corresponding standard deviation.

CpK

The process capability index *CpK* is shown in the column *CpK*. The calculation of the process capability index *CpK* is based on the mean value, the corresponding standard deviation and the upper or lower specification limit. A high value shows that the production lies solidly within the specification limits.

Process location

Graphic display of information on mean value, minimum and maximum value and UTL and LTL.